

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belgaum-590018



MINI PROJECT

REPORT ON

“BloomBalance : A Smart Cycle-Based Health and Lifestyle Management Application for Women”

Submitted in Partial fulfillment of the Requirements for the IV Semester of the Degree of
Bachelor of Engineering in Computer Science & Engineering

By

SHREYA TUMARI (1CR24CS174)

TASMAI S DATH (1CR24CS200)

Under the Guidance of,

Mrs. Paramita Mitra

Asst. Professor, Dept. of CSE



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CMR INSTITUTE OF TECHNOLOGY

#132, AECS LAYOUT, IT PARK ROAD, KUNDALAHALLI, BANGALORE-560037

CMR INSTITUTE OF TECHNOLOGY

#132, AECS LAYOUT, IT PARK ROAD, KUNDALAHALLI, BANGALORE-560037

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

Certified that the project work entitled “BloomBalance : A Smart Cycle-Based Health and Lifestyle Management Application for Women” carried out by **Ms. Shreya Tumari(1CR24CS174)** and **Mr. Tasmai S Dath(1CR24CS200)** are the bonafide students of CMR Institute of Technology, Bengaluru, in partial fulfillment for the award of the degree of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2025-2026. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the departmental library.

The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

Signature of Guide
Mrs. Paramita Mitra
Assistant Professor
Dept. of CSE, CMRIT

Signature of Coordinator
Mrs. Shilpa Shree B
Assistant Professor
Dept. of CSE, CMRIT

Signature of HOD
Dr. Prem Kumar Ramesh
Professor & HOD
Dept. of CSE, CMRIT

DECLARATION

We, the students of 4th semester of Computer Science and Engineering, CMR Institute of Technology, Bangalore declare that the work entitled "BloomBalance : A Smart Cycle-Based Health and Lifestyle Management Application for Women" has been successfully completed under the guidance of Mrs. Paramita Mitra, Assistant Professor, Computer Science and Engineering Department, CMR Institute of technology, Bangalore. This mini project work is submitted in partial fulfillment of the requirements in Computer Science and Engineering during the academic year 2025 - 2026.

Place: Bangalore

Date: 23-05-2026

Team members:

Ms. Shreya Tumari(1CR24CS174)

Mr. Tasmai S Dath(1CR24CS200)

ABSTRACT

BloomBalance is an AI-powered cycle-aware fitness planning application designed to provide personalized workout experiences for women based on menstrual cycle phases, energy levels, symptoms, sleep patterns, and fitness goals. Traditional fitness applications often follow a generalized approach and fail to consider hormonal changes and recovery needs during different phases of the menstrual cycle. BloomBalance addresses this gap by adapting workout recommendations according to the user's physical condition, hormonal changes, and fitness requirements throughout menstrual, follicular, ovulation, and luteal phases.

The application collects user inputs such as age, height, weight, cycle start date, symptoms, sleep hours, energy levels, workout preferences, and fitness goals. Using intelligent recommendation logic, the system generates customized workout plans with suitable exercises, intensity levels, repetitions, sets, and durations. The application also includes progress tracking, workout history, calorie estimation, streak monitoring, analytics visualization, and workout reminders to help users maintain consistency and motivation.

Developed using React Native, Expo, TypeScript, and AsyncStorage, BloomBalance provides a responsive, modern, and user-friendly mobile experience. By combining women's wellness awareness with intelligent fitness planning, the application aims to encourage healthier and more sustainable fitness habits. Future enhancements may include advanced AI recommendations, nutrition guidance, cloud synchronization, wearable integration, and community-based wellness features.

ACKNOWLEDGEMENT

We take this opportunity to express my sincere gratitude and respect to **CMR Institute of Technology, Bengaluru** for providing us a platform to pursue our studies and carry out our final year project

We have a great pleasure in expressing our deep sense of gratitude to **Dr. Sanjay Jain**, Principal, CMRIT, Bangalore, for his constant encouragement.

We would like to thank **Dr. Prem Kumar Ramesh**, HOD, Department of Computer Science and Engineering, CMRIT, Bangalore, who has been a constant support and encouragement throughout the course of this project.

We consider it a privilege and honor to express our sincere gratitude to our guide **Mrs. Paramita Mitra, Asst. Professor**, Department of Computer Science and Engineering, for the valuable guidance throughout the tenure of this review.

We also extend my thanks to all the faculty of Computer Science and Engineering who directly or indirectly encouraged us.

Finally, We would like to thank our parents and friends for all their moral support they have given us during the completion of this work.

TABLE OF CONTENTS

CONTENTS	PAGE NO
Certificate	ii
Declaration	iii
Abstract	iv
Acknowledgement	v
Table of Contents	vi
1. Introduction	1
2. Software Requirements	2
3. Implementation	4
4. Source Code	5
5. Interpretation of Result	8
6. Conclusion and Future Scope	11
7. References	13

LIST OF FIGURES

Figure No.	Figure Caption	Page No.
Fig 3.1	Design Diagram of our solution	4
Fig 5.1	Home screen	8
Fig 5.2	User Input Screen	9
Fig 5.3	Workout Overview screen	10
Fig 5.4	Guided Workout screen	10
Fig 5.5	Progress screen	11
Fig 5.6	User Profile screen	11

LIST OF ABBREVIATIONS

ABBREVIATION	DEFINITION
AI	Artificial Intelligence
API	Application Programming Interface
UI	User Interface
UX	User Experience
VS CODE	Visual Studio Code
NPM	Node Package Manager
HTTP	HyperText Transfer Protocol
JSON	JavaScript Object Notation
Expo	React Native development platform
AsyncStorage	Local storage system in React Native
Firebase	Cloud backend platform
Analytics	User progress and activity analysis
Notifications	Workout reminders and alerts
Cycle Tracking	Monitoring menstrual cycle phases
Adaptive Logic	Dynamic workout adjustment system
Cross-Platform	Supports both Android and iOS